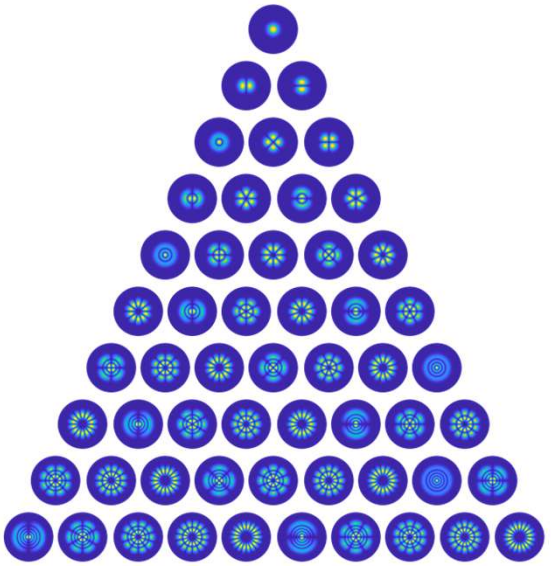
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 1



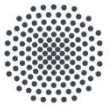
Prof. Dr.-Ing. Georg Rademacher

Organizational matter :

University of Stuttgart
Germany

- This module contains a lecture and an exercise
- Typically, the lecture is held Thursdays at 2 pm in room (V 47-06)
- Exercise is Tuesdays at 2 pm in this room (V 47-06) and will be held by Julian Schneck
- INT offers a lab class „Fachpraktikum (Master) Optische Nachrichtentechnik“ that was newly designed, and I highly recommend attending it!
- There won’t be a course reader (skript) but we will upload the slides from this lectures
- We recommend the following textbook: Govind P. Agrawal, Fiber Optic Communications Systems

Organizational matter:



University of Stuttgart
Germany

- We intend to keep this lecture with lots of interactions
- This includes direct questions but also peer instruction and concept tests
- For peer instruction, we will ask you multiple-choice questions and request for you to discuss those for a few minutes with your neighbors
- Then use your smart phone to vote on the appropriate answer using the given QR code



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Concept Test

Concept test

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What is the major of your studies?

- **A** Smart Information Processing
- **B** Communication Systems
- **C** Electromagnetics Applications
- **D** Smart Systems
- **E** Nano and Opto Electronics
- **F** Power-Electronics Systems
- **G** Electrical Power Systems
- **H** Photonic Engineering (Dedicated)



https://ilias3.uni-stuttgart.de/vote/IX0F

Organizational matter:

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Germany

- There will be approx. 5 graded short tests spread over the lectures of this class
- These will be operated similarly to the concept tests, but you need to log-in with your University of Stuttgart credentials
- You can earn up to 5 bonus points that will count towards the final exam



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Bonus Test-test

Bonus test

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What day is it today?

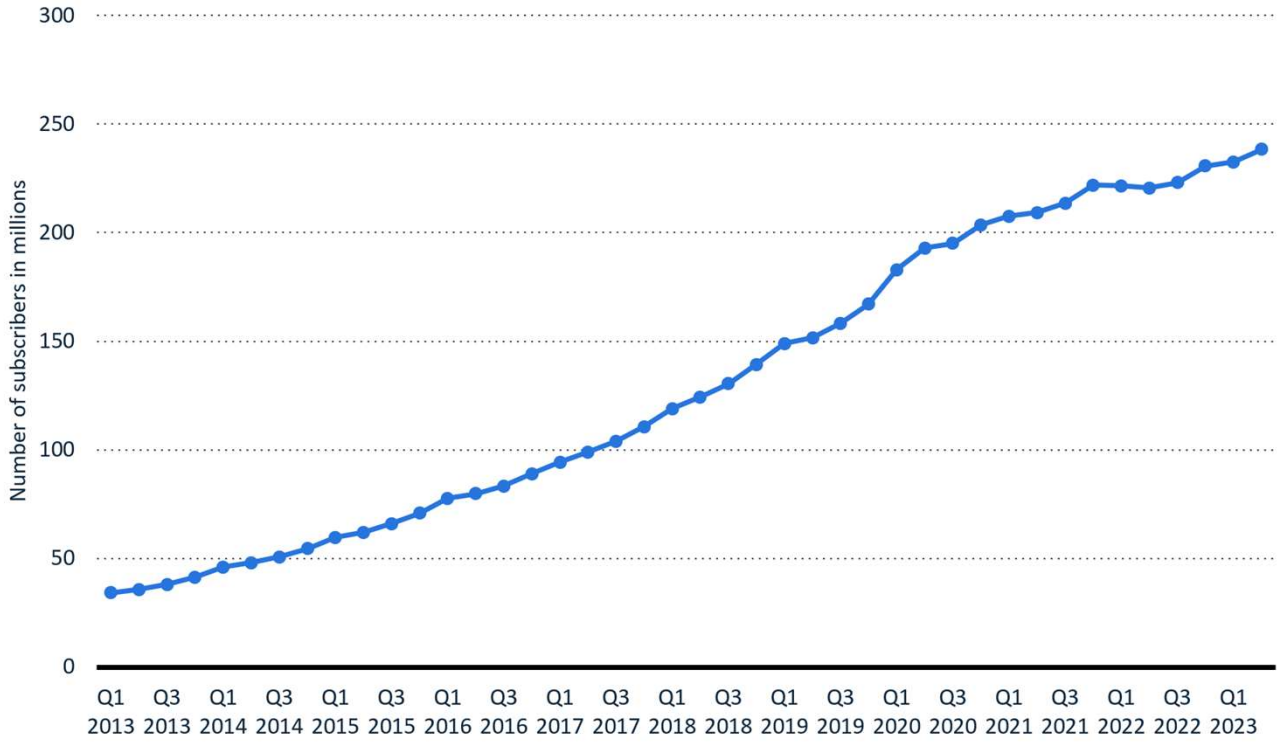
A Monday
B Tuesday
C Wednesday
D Thursday
E Friday
F Saturday
G Sunday

https://ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_MA13


Sei dabei!
emotions-stuttgart.de
emotions
WWW.EMOTIONS-STUTTGART.DE
24.Oktober

Number of Netflix paid subscribers worldwide

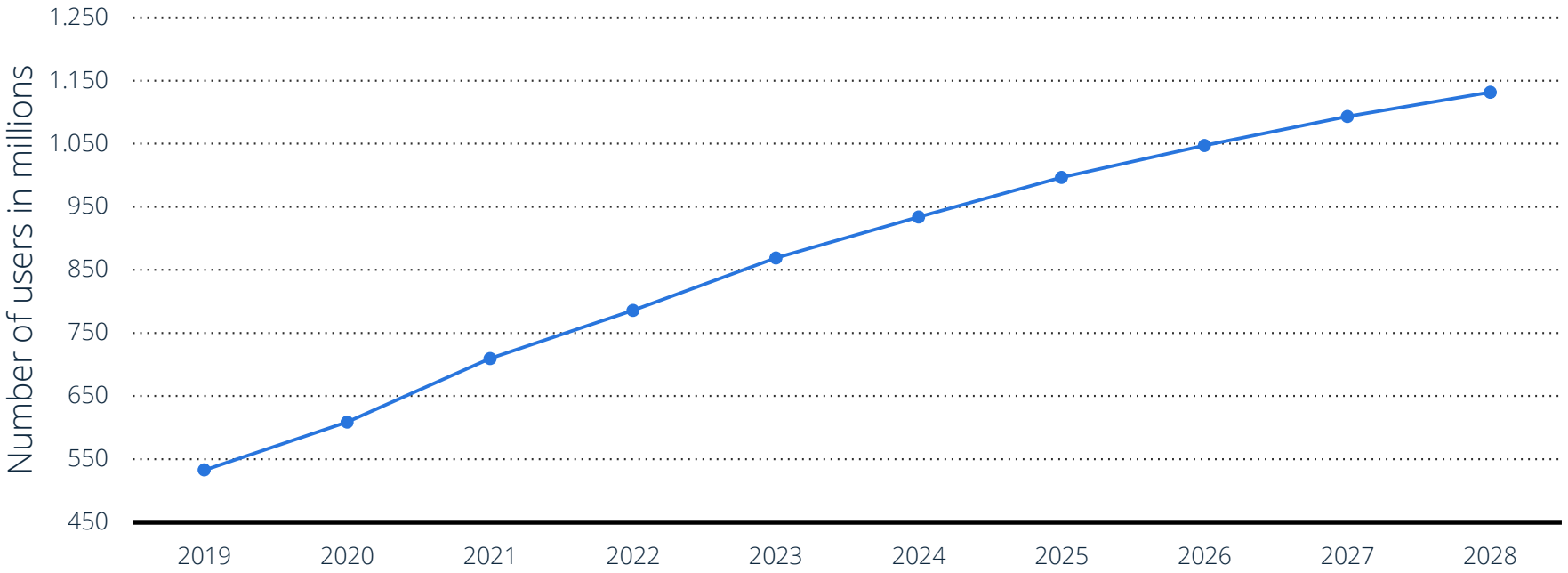
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https://www.statista.com/

Number of YouTube users worldwide from 2019 to 2028

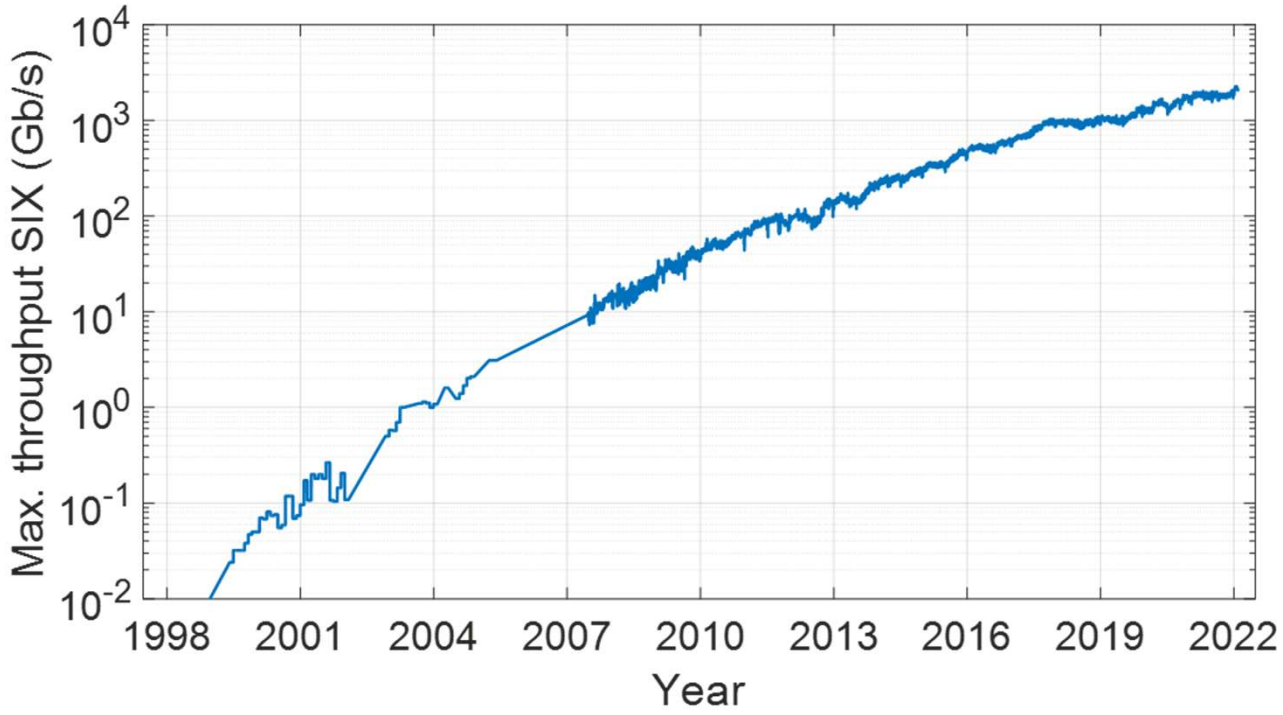
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https://www.statista.com/

Network growth over the past years

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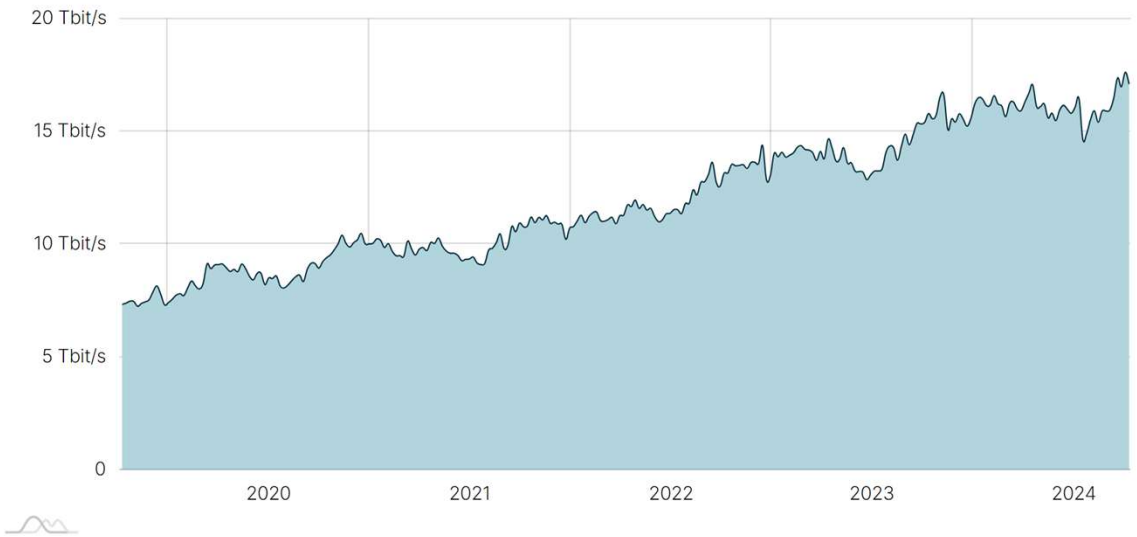


https://www.seattleix.net/

Network growth over the past years

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Germany

Traffic Frankfurt – 5 years



ALL-TIME PEAK
17.65 Tbit/s

GRAPH PEAK
17.65 Tbit/s

GRAPH AVERAGE
8.41 Tbit/s

CURRENT
12.74 Tbit/s

https://www.de-cix.net/en/locations/frankfurt/statistics

Data centers

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Motivation

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Ciena
https://www.ciena.com › ... › Newsroom › Press Releases

Ciena to Acquire Nubis Communications to Expand its ...

22 Sept 2025 — Under the terms of the agreement, **Ciena will acquire Nubis in an all-cash transaction for $270 million**, subject to a customary purchase price ...

NVIDIA Corp
ETR: NVD

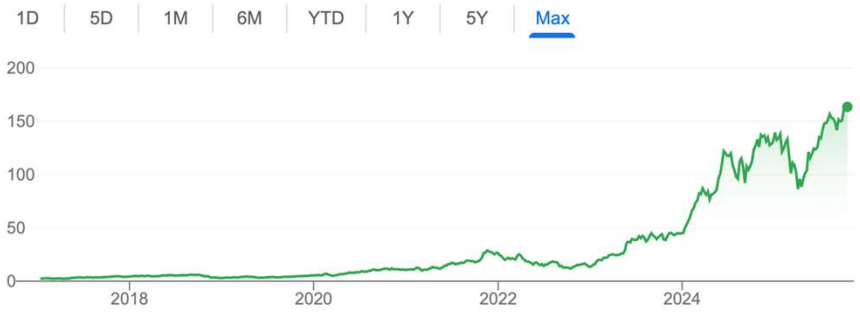
Market Summary > NVIDIA Corp

163,60 EUR

+161.16 (6,604.92%) ↑ all time

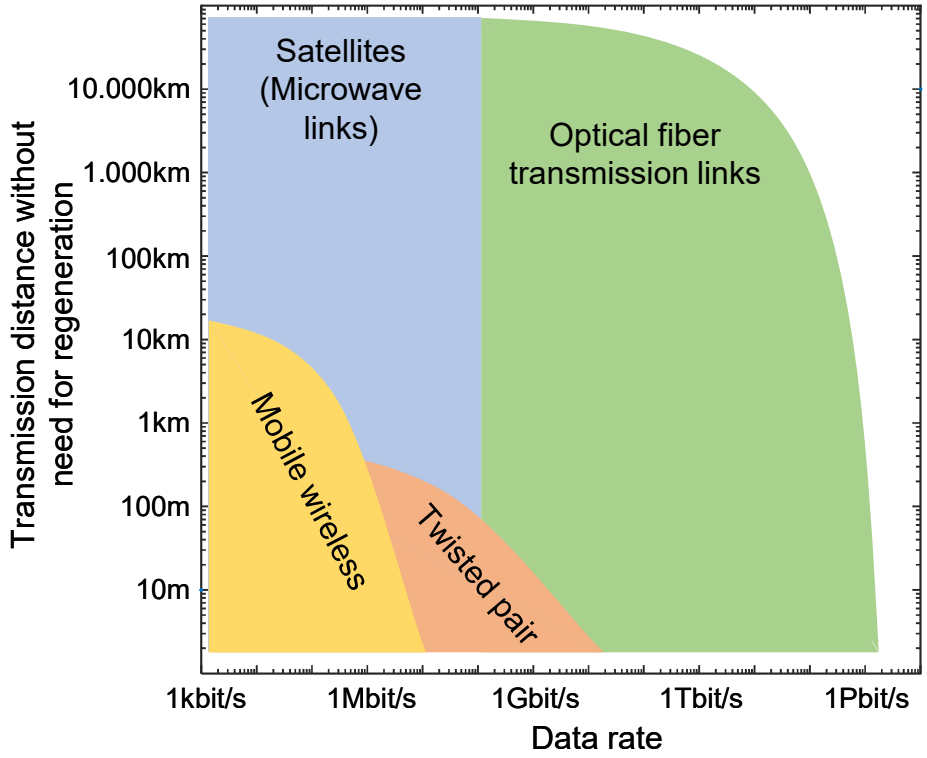
13 Oct, 11:35 CEST • Disclaimer

+ Follow



Communication links

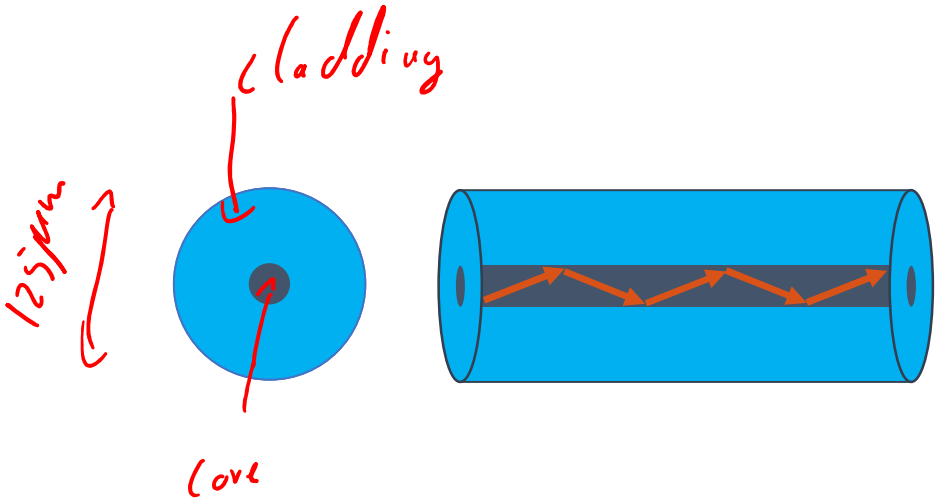
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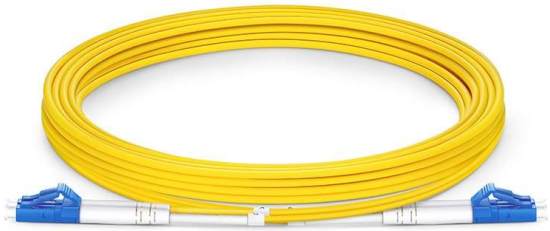
P. Winzer et al., Advanced optical modulation formats, Proceeding of the IEEE vol 94, pp. 952 - 985 (2006)

What is an optical fiber?

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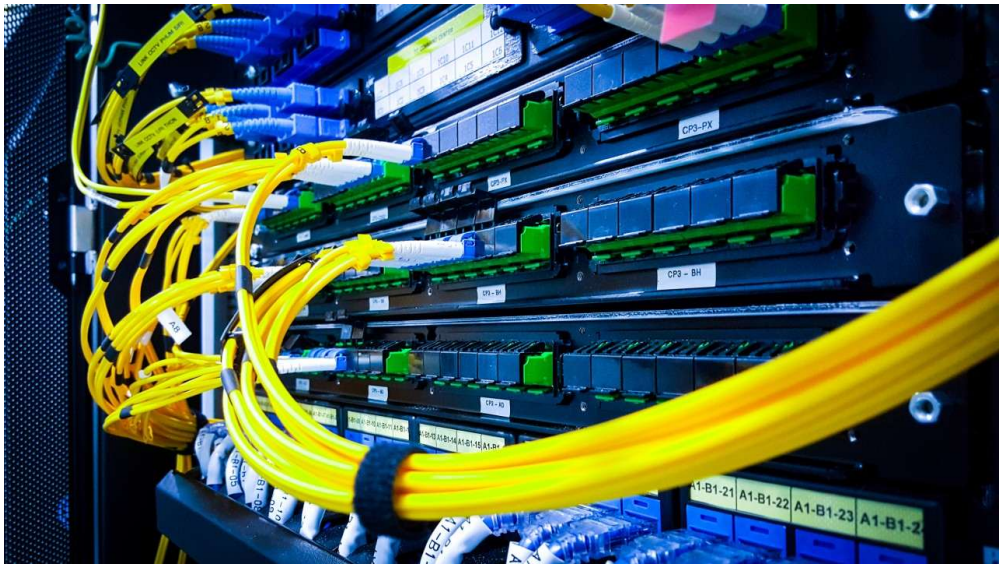


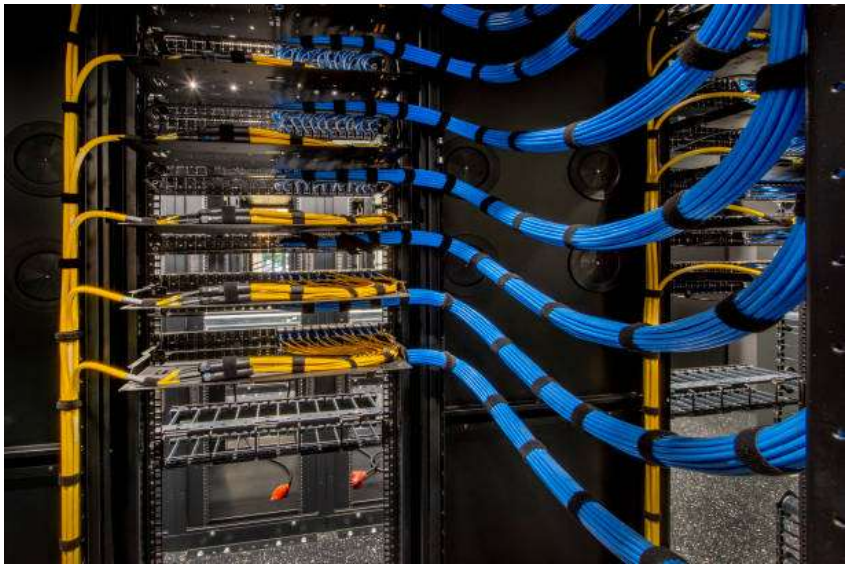




Inside a data center

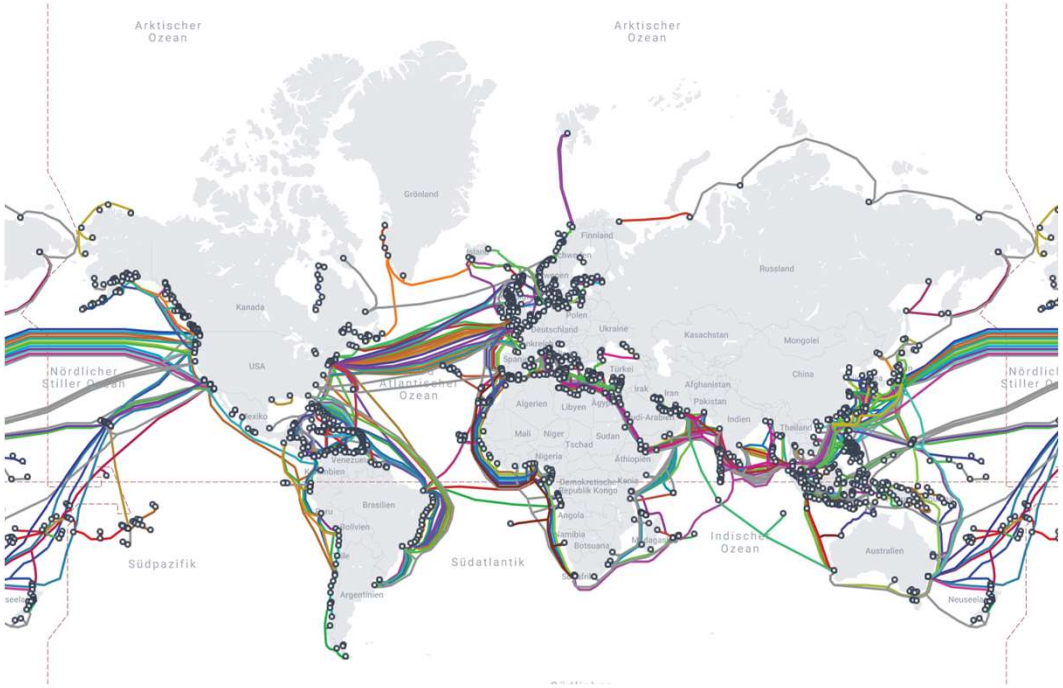
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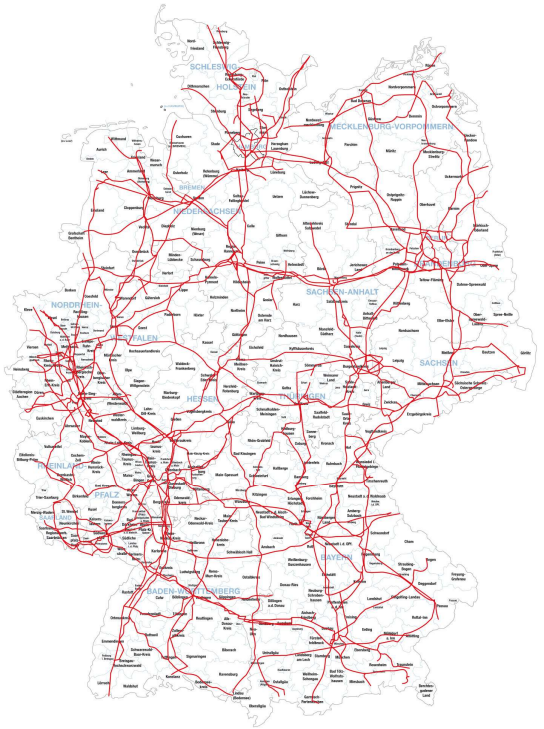




Global information infrastructure

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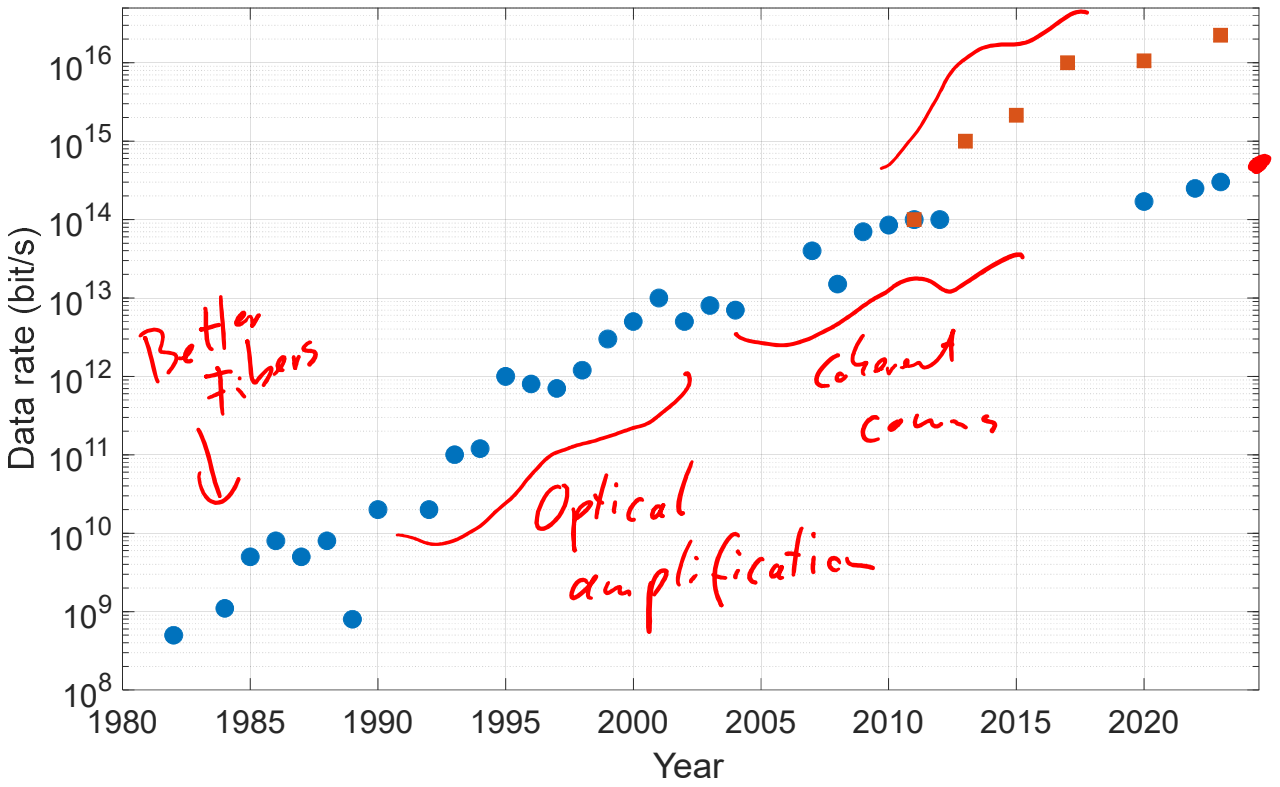




https://www.submarinecablemap.com/

https://www.1und1.net/business/kompetenz/glasfaser/unser-netz

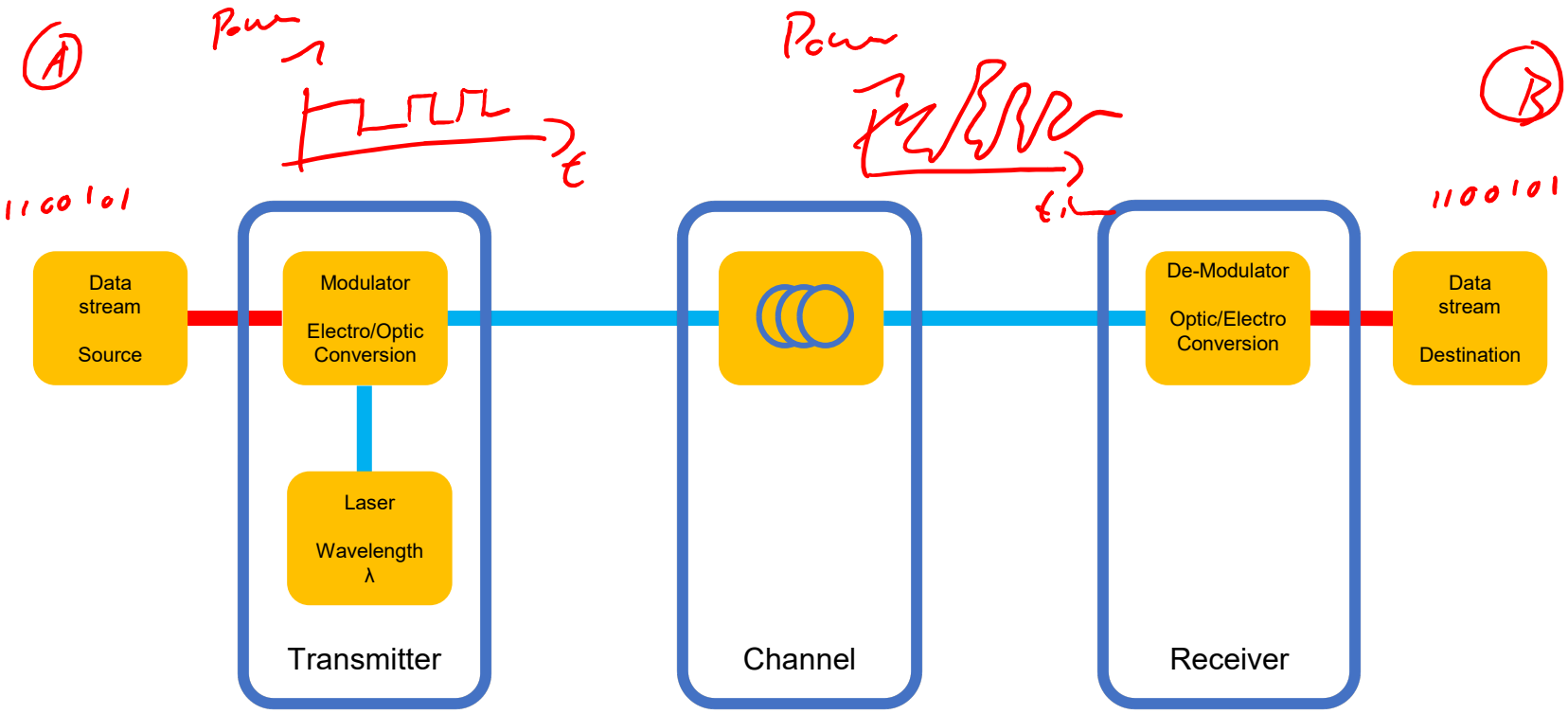
Data-rate records in a single optical fiber



B.J. Puttnam, G. Rademacher, R.S. Luis. Space-Division Multiplexing for Optical Fiber Communications. Optica, vol. 8, no. 9, 2021

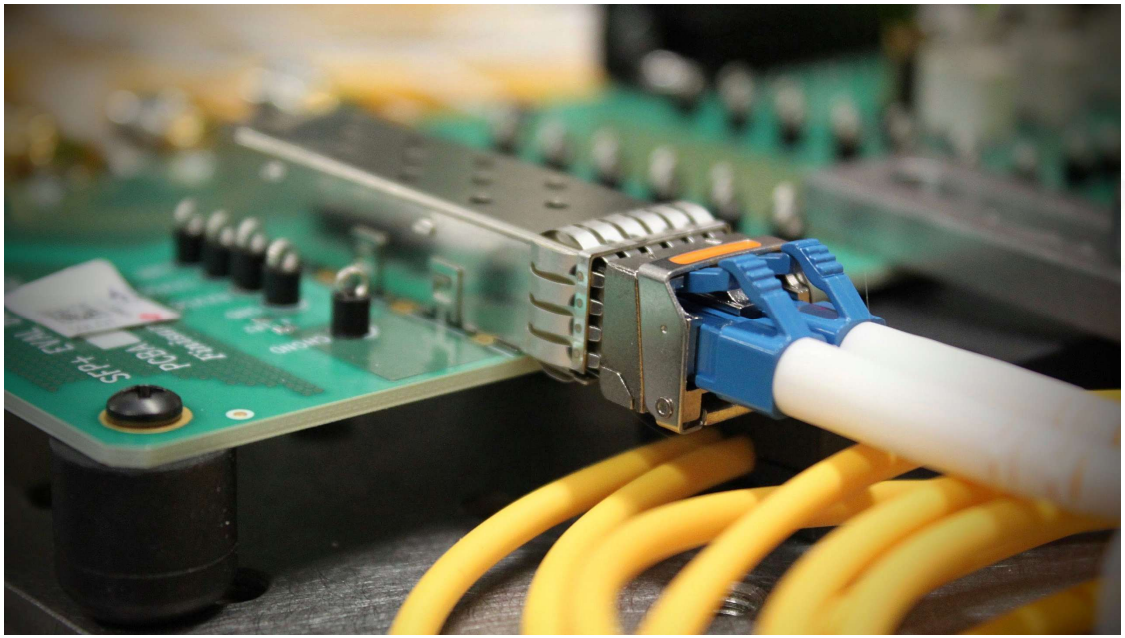
Principle of an optical fiber transmission link

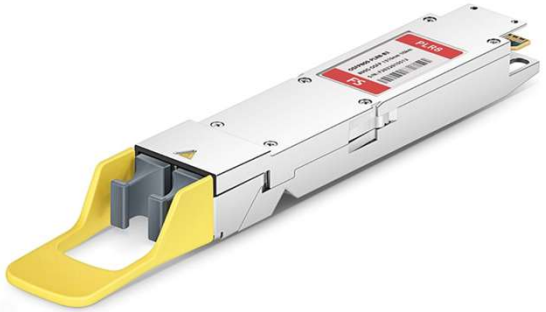
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How does this look in practice?

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Small form-factor plugable (SFP), here 400 Gbit/s Ethernet

Inside the lab

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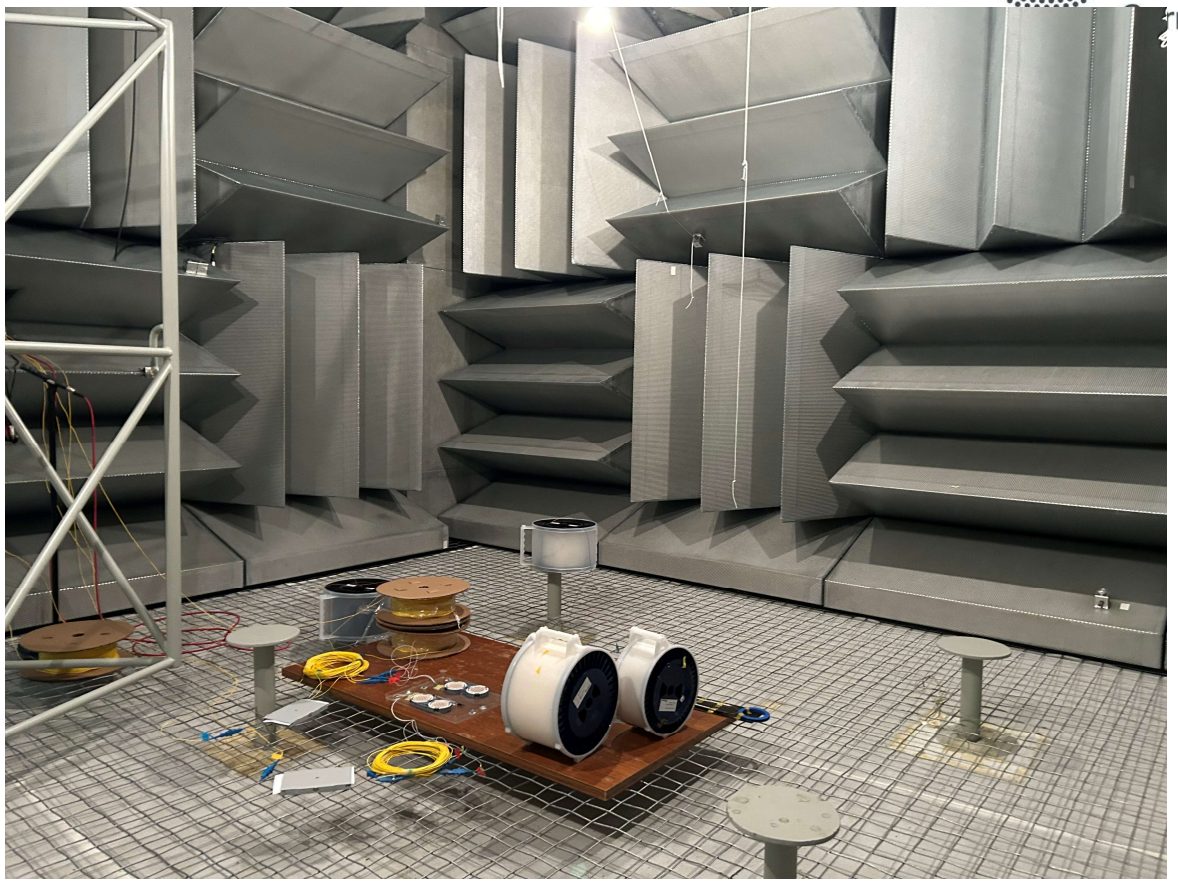
NICT

INT Lab





Inside the lab



Why do we like optical fibers for transmission?

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1. Security
2. Large Bandwidth → high data rates
3. low attenuation
4. cheap
5. low EMI
6. high density cables



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Concept Test

Concept test

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Assume a communication system operates at a bit rate up to 10% of the carrier frequency. How many 1080p HD streams at 5 Mb/s can be transmitted with an optical carrier compared to a microwave carrier at 5 GHz?

- **A** 3.87 billion
- **B** 38700
- **C** 3.87 million

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(Tentative) Content of this course:

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Basics

a. Planewaves
b. Chromatic dispersion
c. Sellmeier equation for refractive index
d. Attenuation
e. Reflection, snells law

Optical Fibers

a. Step index fiber (MMF, SMF), fiber modes
b. Graded index fiber
c. Multi core fibers (strongly coupled, weakly coupled)
d. Hollow-Core fibers?
e. Coupling fibers

Optical amplifiers

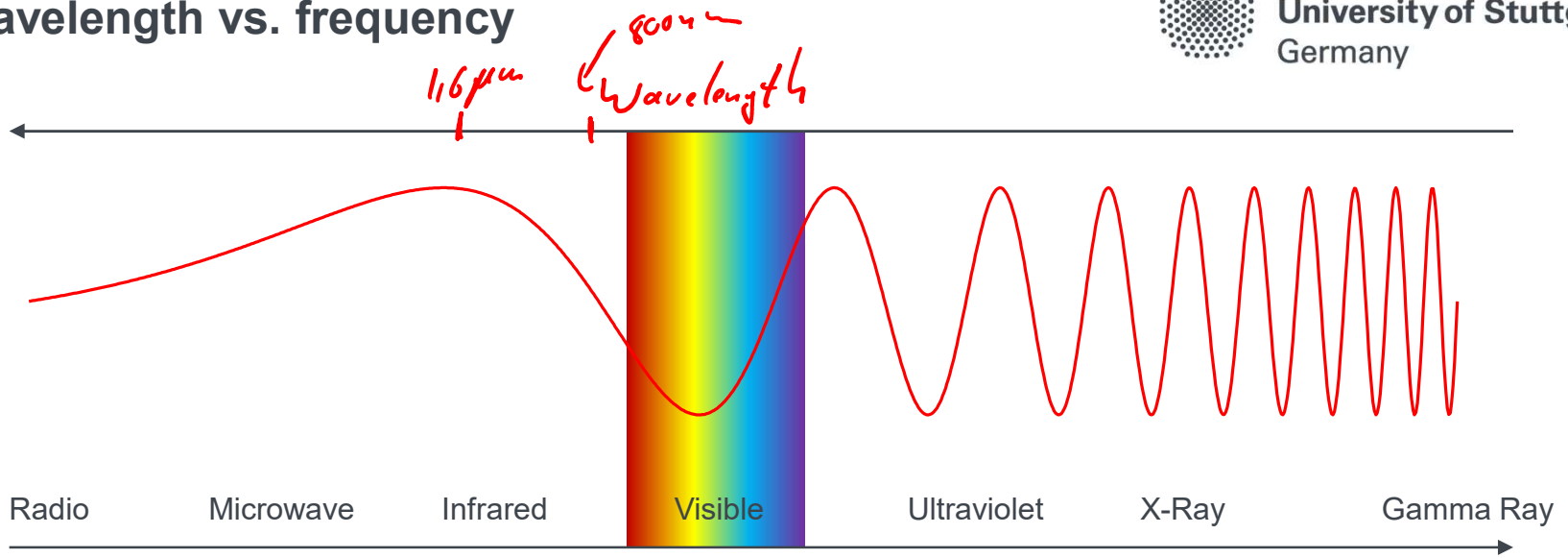
a. Semiconductor optical amplifiers (SOAs)
b. Erbium-doped fiber amplifiers (EDFAs)
c. Noise

Optical transmission systems

a. SMF based systems, noise analysis, SNR analysis, BER
b. Nonlinear signal distortions
c. SDM transmission systems based on MCF, MMF

Wavelength vs. frequency

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λ = Wavelength (free-space)

c = speed of light (free-space) $\sim 3 \cdot 10^8 \frac{m}{s}$

f: frequency

$$\lambda = \frac{c}{f}$$

Plane waves

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$\varepsilon = \varepsilon_0 \cdot \varepsilon_r \quad , \quad \varepsilon_r = n^2 \qquad \omega = 2 \cdot \pi \cdot f$

$\vec{E}(t) = \mathrm{Re}\{ \underline{\vec{E}} \cdot \exp(j \omega t) \} \quad , \quad \vec{H}(t) = \mathrm{Re}\{ \underline{\vec{H}} \cdot \exp(j \omega t) \}$

$\nabla \times \underline{\vec{E}} = -j \omega \mu_0 \underline{\vec{H}}$

$\nabla \times \underline{\vec{H}} = j \omega \varepsilon \underline{\vec{E}} = j \omega \varepsilon_0 n^2 \underline{\vec{E}}$

Plane waves

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$$\nabla \times (\nabla \times \underline{\vec{E}}) - k_0^2 n^2 \underline{\vec{E}} = 0 \qquad , \quad k_0 = \omega \sqrt{\mu_0 \cdot \varepsilon_0}$$

$$\nabla \times (\nabla \times \underline{\vec{E}}) = \nabla \underbrace{(\nabla \underline{\vec{E}})}_{=0} - \Delta^2 \underline{\vec{E}} = -\nabla^2 \underline{\vec{E}}$$

Plane waves

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$$\nabla^2 \underline{\vec{E}} + k_0^2 n^2 \underline{\vec{E}} = 0$$

$$\nabla^2 \underline{\vec{H}} + k_0^2 n^2 \underline{\vec{H}} = 0$$

∇^2 : Laplace operator

$$\rightarrow \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\underline{\vec{E}} = \underline{\vec{E}}_0 \exp\left(-j \cdot \vec{k} \cdot \vec{r}\right)$$

$$= \underline{\vec{E}}_0 \exp\left(-j\left(k_x x + k_y y + k_z z\right)\right)$$

$$\vec{k} = \begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix} : \text{wave vector}$$

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\vec{k} \cdot \vec{k} = \left(\vec{k}\right)^2 = k_x^2 + k_y^2 + k_z^2 = k_0^2 n^2$$

Plane waves

planes with constant phase: $\delta = \vec{k} \cdot \vec{r}$

$$\underline{\vec{H}} = \frac{\vec{k} \times \underline{\vec{E}}}{\omega \mu_0} = \underline{\vec{H}_0} \cdot \exp(-j \cdot \vec{k} \cdot \vec{r})$$

$$\underline{\vec{H}_0} = \frac{\vec{k} \times \underline{\vec{E}_0}}{\omega \mu_0}$$

Poynting-Vector $\underline{\vec{S}}$

$$\underline{\vec{S}} = \frac{1}{2} (\underline{\vec{E}} \times \underline{\vec{H}})$$

energy flux



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Concept Test

Concept test

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What is the unit of the Poynting vector $\vec{S}$?

A $\frac{\mathrm{W}}{\mathrm{m}^3}$

B $\frac{\mathrm{W}}{\mathrm{m}^2}$

C $\frac{\mathrm{J}}{\mathrm{m}^2}$

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University of Stuttgart
Institute of electrical and optical communications

Thank you!



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